

Labour Productivity and Comparative Advantage: The Ricardian Model

ECON 308 – International Economic Relations Chapter 3

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Fall 2026

Opening Puzzle: How Can Bangladesh Compete?

- Bangladesh has lower labour productivity than the US in **every single industry**: garments, electronics, agriculture, steel.
- Yet Bangladesh is one of the world's **largest garment exporters** (\$34 billion in 2019), while the US imports Bangladeshi clothing by the billions.
- The US is absolutely more productive in *everything*. So why does trade happen?
- **David Ricardo's answer (1817)**: absolute productivity does not matter for trade. What matters is *relative* productivity — comparative advantage.

Ricardo's Insight

A country should export the good in which it has the *lowest opportunity cost* of production, even if it is absolutely less efficient in all goods.

This is one of the most profound and counterintuitive results in all of economics.

Learning Objectives

After this lecture you will be able to:

1. Distinguish **absolute advantage** from **comparative advantage** and explain why only the latter determines trade patterns.
2. Define unit labour requirements and calculate opportunity costs.
3. Derive each country's **production possibility frontier** and autarky equilibrium.
4. State the condition for trade and determine the **pattern of specialisation**.
5. Show that **both** countries gain from trade by reaching consumption beyond their PPFs.
6. Explain how wages and the terms of trade are determined under free trade.
7. Evaluate empirical evidence on the Ricardian model.

Model Setup and Assumptions

The Ricardian model uses the simplest possible framework:

- **2 countries:** Home (H) and Foreign (F^*).
- **2 goods:** Cloth (C) and Wine (W) — or any two tradable goods.
- **1 factor of production:** Labour (L) only.
- Labour is **perfectly mobile** between sectors within a country.
- Labour is **immobile** between countries.
- **Perfect competition:** price equals unit cost.
- **Constant returns to scale:** double inputs, double output.
- **Full employment:** all labour is used.

Why one factor?

Using only labour isolates the role of *technology* in determining comparative advantage. The single-factor assumption is relaxed in Chapters 4–5 (specific factors, H-O), where capital and land are introduced.

What this model cannot do:

- Explain income distribution within countries.
- Account for capital or land.
- Explain intra-industry trade.

Key Variables Defined

Unit labour requirements (ULRs):

a_{LC} = hours of labour needed to produce 1 unit of cloth (Home)

a_{LW} = hours of labour needed to produce 1 unit of wine (Home)

a_{LC}^* = ULR for cloth in Foreign

a_{LW}^* = ULR for wine in Foreign

Lower ULR $\Rightarrow$ higher labour productivity.

$1/a_{LC}$ = output of cloth per worker-hour (Home productivity)

Other notation:

L = total labour supply (Home)

L^* = total labour supply (Foreign)

Q_C, Q_W = quantities of cloth and wine produced (Home)

w = wage rate (Home), w^* = wage rate (Foreign)

P_C, P_W = prices of cloth and wine

$p \equiv P_C/P_W$ = relative price of cloth (the terms of trade)

Prices Under Competition

$P_C = w \cdot a_{LC}$ and $P_W = w \cdot a_{LW}$

Price = unit labour cost.

Absolute vs. Comparative Advantage: The Crucial Distinction

Absolute Advantage

Country A has absolute advantage in good X if it uses *fewer labour hours* per unit than Country B:

$$a_{LX} < a_{LX}^*$$

A is simply more efficient at producing X.

Comparative Advantage

Country A has comparative advantage in X if its *opportunity cost* of producing X is lower:

$$\frac{a_{LC}}{a_{LW}} < \frac{a_{LC}^*}{a_{LW}^*}$$

Ricardo's Profound Insight

Even if Country A is absolutely better at *everything*, trade is still mutually beneficial if the two countries have different *opportunity costs*.

Why? Because what matters for specialisation is not how much labour you use — it is what you *give up* to produce one more unit.

The Opportunity Cost Formula

How to compute opportunity cost (OC):

The OC of producing one unit of cloth in Home is the amount of wine that could have been produced with those same labour hours:

$$\text{OC of 1 cloth (Home)} = \frac{a_{LC}}{a_{LW}} \text{ units of wine}$$

Because: producing 1 unit of cloth requires a_{LC} hours; those hours could produce a_{LC}/a_{LW} units of wine.

Autarky (no-trade) relative price:

Under competition: $P_C/P_W = a_{LC}/a_{LW}$

The relative price equals the opportunity cost.

Step-by-step: OC of cloth for Home

1. 1 unit of cloth requires a_{LC} hours.
2. Those a_{LC} hours could produce a_{LC}/a_{LW} units of wine.
3. OC of cloth = a_{LC}/a_{LW} wine.

Step-by-step: OC of wine for Home

1. 1 unit of wine requires a_{LW} hours.
2. Those hours could produce a_{LW}/a_{LC} units of cloth.
3. OC of wine = a_{LW}/a_{LC} cloth.

Numerical Example: Home and Foreign

	a_{LC}	a_{LW}	OC of cloth	OC of wine
Home	1 hr	2 hrs	$\frac{1}{2}$ wine	2 cloth
Foreign	6 hrs	3 hrs	2 wine	$\frac{1}{2}$ cloth

Step 1 — Absolute advantage:

- Cloth: Home uses 1 hr vs. Foreign 6 hrs $\Rightarrow$ Home has absolute advantage.
- Wine: Home uses 2 hrs vs. Foreign 3 hrs $\Rightarrow$ Home has absolute advantage.
- Home is *absolutely more productive in both goods*.

Step 2 — Comparative advantage:

- Home OC of cloth: $1/2$ wine.
- Foreign OC of cloth: $6/3 = 2$ wine.
- $1/2 < 2 \Rightarrow$ **Home has CA in cloth.**
- Home OC of wine: $2/1 = 2$ cloth.
- Foreign OC of wine: $3/6 = 1/2$ cloth.
- $1/2 < 2 \Rightarrow$ **Foreign has CA in wine.**

Result

Each country specialises in the good where its OC is *lower*.

Home's Production Possibility Frontier

Home's resource constraint:

$$a_{LC} Q_C + a_{LW} Q_W = L$$

Variables:

Q_C, Q_W = cloth and wine outputs

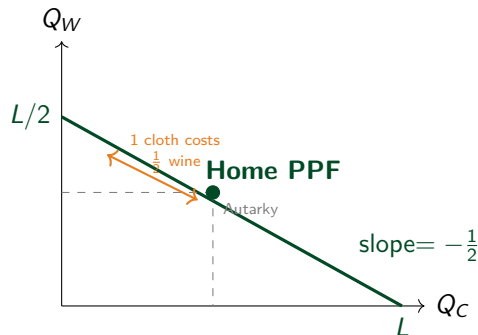
L = total labour endowment (fixed)

$a_{LC} = 1, a_{LW} = 2$ (from example)

Intercepts:

- Max cloth: $L/a_{LC} = L$ (all labour in cloth).
- Max wine: $L/a_{LW} = L/2$ (all labour in wine).

Slope: $-a_{LC}/a_{LW} = -1/2$
(OC of 1 cloth = 1/2 unit of wine)



Reading the PPF: What the Slope Tells Us

The PPF slope is the opportunity cost:

- **Slope** = $-a_{LC}/a_{LW} = -1/2$
- Moving right by 1 unit of cloth requires moving down by $1/2$ unit of wine.
- This $1/2$ unit of wine is the *opportunity cost* of cloth.
- Under autarky, competitive equilibrium is on the PPF at slope = $-(P_C/P_W)^{\text{autarky}}$.
- The PPF is a straight line (constant OC) because there is only one factor (labour) and constant returns.

Why a Straight Line?

With one factor (labour) and constant returns, every unit of cloth costs exactly a_{LC}/a_{LW} units of wine — regardless of how much cloth is already being produced.

In Chapters 4–5 (multiple factors), the PPF is *bowed out* because factors differ in their suitability for each good.

Chapter 5 vs. Chapter 3 PPF:

Ricardian: straight line (OC constant)
H-O: bowed out (OC rising)

Foreign's Production Possibility Frontier

Foreign's resource constraint:

$$a_{LC}^* Q_C^* + a_{LW}^* Q_W^* = L^*$$

$a_{LC}^* = 6$, $a_{LW}^* = 3$ (from example)

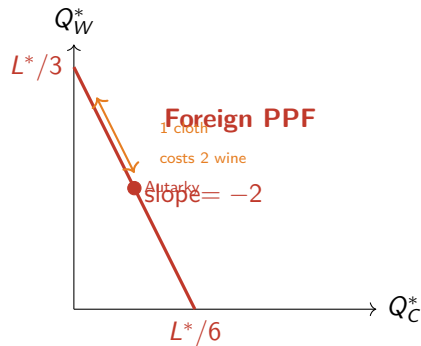
Intercepts:

- Max cloth: $L^*/6$
- Max wine: $L^*/3$

Slope: $-a_{LC}^*/a_{LW}^* = -6/3 = -2$
(OC of 1 cloth = 2 units of wine)

Autarky price: $P_C^*/P_W^* = 2$

Foreign's cloth is *more expensive* (higher OC)
than Home's cloth.



Comparing the Two Countries

Country	a_{LC}	a_{LW}	OC of cloth	OC of wine	Comparative adv.
Home	1	2	$\frac{1}{2}$ wine	2 cloth	Cloth
Foreign	6	3	2 wine	$\frac{1}{2}$ cloth	Wine

Key observations:

1. Home's autarky price of cloth:
 $P_C/P_W = 1/2$.
2. Foreign's autarky price of cloth:
 $P_C^*/P_W^* = 2$.
3. Because $1/2 < 2$, cloth is *relatively cheaper* in Home under autarky.
4. A potential gain from trade exists: Foreign consumers can get cloth cheaper by importing from Home; Home consumers can

The No-Trade Impossibility

If countries do not trade, each is stuck on its own PPF. Opening trade allows both to consume *beyond* their PPFs by exploiting the difference in opportunity costs.

The Pattern of Trade: Theorem and Condition

Ricardian Trade Theorem

Under free trade, each country will **completely specialise** in the good where it has comparative advantage, provided the world relative price $p \equiv P_C/P_W$ satisfies:

$$\frac{a_{LC}}{a_{LW}} < p < \frac{a_{LC}^*}{a_{LW}^*}$$

Interpretation:

- If $p > a_{LC}/a_{LW}$: Home earns more from cloth than from wine per unit of labour $\Rightarrow$ specialises fully in cloth.
- If $p < a_{LC}^*/a_{LW}^*$: Foreign earns less from cloth per unit of labour $\Rightarrow$ specialises fully in wine.
- For our example: $1/2 < p < 2$ ensures

Special cases:

$p = a_{LC}/a_{LW}$: Home is indifferent; production is indeterminate.

$p = a_{LC}^*/a_{LW}^*$: Foreign is indifferent.

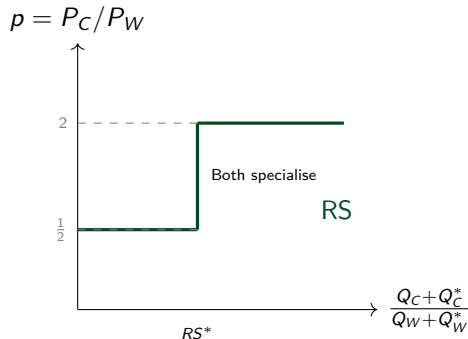
p outside the range: one country specialises fully; the other partially.

World Relative Supply Curve

World RS curve plots $(Q_C + Q_C^*)/(Q_W + Q_W^*)$ as a function of relative price p :

- If $p < a_{LC}/a_{LW} = 1/2$: both produce wine only $\Rightarrow RS = 0$.
- If $p = 1/2$: Home is indifferent; RS jumps from 0 to $L/a_{LC}^* \div L^*/a_{LW}^*$.
- If $1/2 < p < 2$: Home fully in cloth, Foreign fully in wine $\Rightarrow RS = (L/a_{LC})/(L^*/a_{LW}^*)$ — flat.
- If $p = 2$: Foreign is indifferent; RS jumps again.
- If $p > 2$: both in cloth $\Rightarrow RS = \infty$.

RS is a step function.



World Relative Demand and Equilibrium

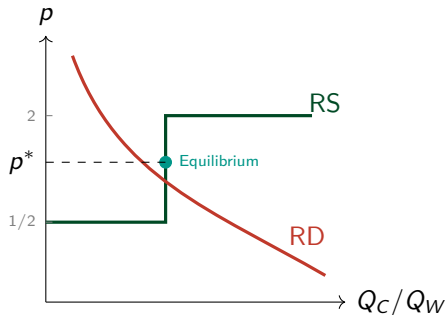
World Relative Demand (RD):

downward-sloping.

Higher $p = P_C/P_W \Rightarrow$ consumers worldwide substitute toward wine $\Rightarrow$ less cloth demanded relative to wine.

World equilibrium: where RD crosses RS.

- If RD crosses the flat part of RS: p^* is between $1/2$ and $2 \Rightarrow$ both countries specialise.
- If RD crosses a vertical part: one country is at the "kink" and produces both goods (indeterminate specialisation).



Gains from Trade: Home

Under autarky:

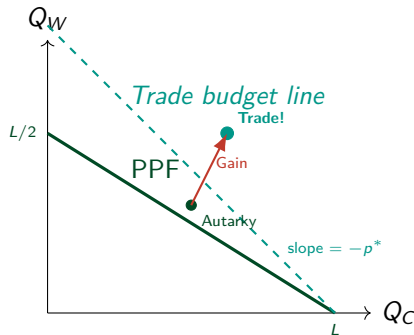
Home consumes on PPF at price ratio

$$P_C/P_W = 1/2.$$

Under free trade at $p^* = 1$ (example):

1. Home specialises fully in cloth, producing $L/a_{LC} = L$ units.
2. Home *sells cloth* and *buys wine* at world price $p^* = 1$.
3. Budget line slope is $-p^* = -1$ (steeper than PPF slope $-1/2$).
4. Home can reach an indifference curve **above** the autarky level.

The gain: Home now buys wine at $p^* = 1$ cloth/wine, but can produce wine only at a cost of 2 cloth/wine domestically. Trade saves



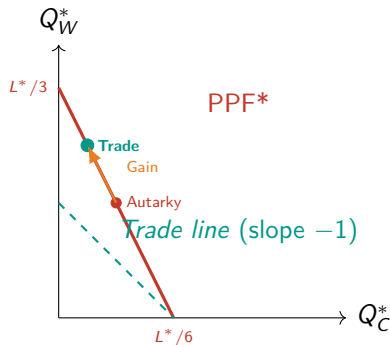
Gains from Trade: Foreign

Foreign under autarky: produces at

$$P_C^*/P_W^* = 2.$$

Foreign under free trade at $p^* = 1 < 2$:

1. Foreign specialises in wine (where its OC is low: 1/2 cloth).
2. Sells wine and buys cloth at world price $p^* = 1$.
3. $p^* = 1 < 2 =$ Foreign's autarky price $\Rightarrow$ cloth is *cheaper* from trade.
4. Budget line slope = -1 (flatter than Foreign's PPF slope -2).
5. Foreign reaches a higher consumption level.



Gain: Foreign can now get cloth at 1 unit of wine per cloth, vs. 2 units of wine it would cost

The Terms of Trade: Definition and Determination

Definition

The **terms of trade (ToT)** for Home = $P_C/P_W = p^*$ (price of its export in terms of its import).

The ToT must satisfy: $a_{LC}/a_{LW} < p^* < a_{LC}^*/a_{LW}^*$ (both autarky prices bound the ToT).

Higher p^* (ToT) benefits Home:

- Each unit of cloth exported buys more wine.
- Home consumes more wine.

Lower p^* benefits Foreign:

- Each unit of wine buys more cloth.

The exact ToT is where world RS = world RD.

Canada's Terms of Trade 2002–2014

Rising oil and commodity prices dramatically improved Canada's ToT. Canada's export prices rose while import prices (manufactures) fell. Real national income rose $\sim 15\%$ more than real GDP — a pure ToT gain.

After 2014: oil price collapse worsened ToT sharply.

Wages and the Distribution of Gains

Home's wage under trade:

Home specialises in cloth; workers earn:

$$w = \frac{P_C}{a_{LC}} = \frac{p^* \cdot P_W}{a_{LC}}$$

Foreign's wage:

$$w^* = \frac{P_W}{a_{LW}^*}$$

Relative wage:

$$\frac{w}{w^*} = \frac{p^* \cdot P_W / a_{LC}}{P_W / a_{LW}^*} = p^* \cdot \frac{a_{LW}^*}{a_{LC}}$$

- More productive country earns higher wages.

Key Result

Higher productivity $\Rightarrow$ higher wages — but this does not prevent trade. Each country still benefits from the exchange even though wages differ.

What if wages cannot equate?

If workers cannot move between sectors, the gains from trade are unequally distributed — studied in Chapter 4 (specific factors). In the Ricardian model, all workers earn the same wage within a country.

Relative Wage Bounds

Where must w/w^* lie?

For trade to occur (both countries to specialise), the wage ratio must satisfy:

$$\frac{a_{LC}^*}{a_{LC}} > \frac{w}{w^*} > \frac{a_{LW}^*}{a_{LW}}$$

In our example: $a_{LC}^*/a_{LC} = 6$ and $a_{LW}^*/a_{LW} = 3/2$:

$$6 > \frac{w}{w^*} > \frac{3}{2}$$

Home's wage must be between $1.5\times$ and $6\times$ Foreign's wage.

- If $w/w^* = 6$: Foreign is indifferent in cloth.
- If $w/w^* = 3/2$: Foreign is indifferent in

Economic Meaning

Even though Home is absolutely more productive (by factor $3-6\times$), it cannot charge wages so high that it loses comparative advantage in cloth.

Wages adjust endogenously to ensure each country *remains competitive* in the good where it has comparative advantage.

Productivity Differences and Wages: Real-World Data

Do real wages reflect productivity?

The Ricardian model predicts: $w \propto$ labour productivity.

Country	Wage (USD/hr)	Productivity (USA=100)	Wage/Prod. ratio
USA	35	100	35%
Germany	43	108	40%
South Korea	22	67	33%
Mexico	5	28	18%
Bangladesh	0.6	4	15%

Source: BLS, ILO, World Bank, 2019

- High-productivity countries have high wages (Ricardian prediction confirmed).
- The wage/productivity ratio is not constant: richer countries pay more per unit of productivity — reflecting non-traded services, institutional factors.
- Bangladesh's low wages reflect low productivity — not a threat to US wages in high-productivity sectors.

Verdict

Wages track productivity, not just absolute wages. Trade with low-wage countries does not impoverish high-wage countries.

Empirical Evidence: The MacDougall Test

MacDougall (1951): tested the Ricardian model using 1937 data.

Test: In industries where the US had a *larger* productivity advantage over the UK, did the US export more?

Finding: Yes — a strong positive correlation between US productivity advantage and US export share.

Industry	US/UK prod.	US wins?
Pig iron	3.0×	Yes
Motor vehicles	4.0×	Yes
Cigarettes	2.0×	Tie
Wool cloth	1.5×	No

Source: MacDougall (1951)

- **Balassa (1963):** confirmed for post-WWII data.
- **Golub & Hsieh (2000):** US–Asia trade strongly correlated with unit labour cost differences.
- **Conclusion:** the Ricardian model's prediction about the *direction* of trade is empirically well-supported.

But...

Ricardian model predicts *complete* specialisation — rarely observed. Real countries partially specialise. Chapters 4–5 explain why.

Bangladesh vs. China: Comparative Advantage in Action

Both export garments. How?

- China has higher absolute productivity in *both* garments and electronics.
- But China's **opportunity cost** of garments is rising: its comparative advantage is shifting toward electronics, machinery, and AI-enabled products.
- Bangladesh's opportunity cost of garments remains low — it lacks a competitive electronics sector.
- Result: China moves up the value chain; Bangladesh stays in garments.

	Bangladesh	China
Wage (mfg, USD/hr)	0.57	4.99
Garment exports	\$34B	\$185B
Electronics exports	\$0.4B	\$842B
Garments/Electronics	85×	0.22×

Source: ITC, World Bank, 2019

Ricardian Prediction Confirmed

Each country exports where its *relative* cost is lowest, not where its absolute cost is lowest.

Limitations of the Ricardian Model and What Comes Next

What the Ricardian model does well:

- Explains the *direction* of trade (comparative advantage).
- Shows mutual gains from trade for both countries.
- Connects productivity to wages (empirically supported).
- Generates clear, testable predictions.

What it misses:

- **Complete specialisation:** not observed in practice.
- **One factor:** cannot explain income distribution effects of trade.
- **No capital or land:** ignores factor

Building Up the Model

- **Ch. 4:** Add capital and land as specific factors. Show trade creates *winners and losers* within a country.
- **Ch. 5:** Long run: all factors mobile. H-O model.
- **Ch. 7:** Add economies of scale. Explain intra-industry trade.

Key Takeaways

1. **Comparative advantage** (not absolute advantage) determines the pattern of trade. A country exports the good where its opportunity cost is *relatively* lowest.
2. The **PPF** is linear in the Ricardian model (constant OC). The slope equals the autarky relative price.
3. Under free trade, each country **completely specialises** in its comparative advantage good.
4. Both countries gain: trade allows consumption **beyond the PPF**.
5. The **terms of trade** must lie between the two autarky prices for trade to be mutually beneficial.
6. **Wages reflect productivity**: more productive countries pay higher wages, but all countries trade.
7. Empirical evidence **strongly supports** the Ricardian pattern of trade.

Next Lecture: Chapter 4 — Specific Factors Model

What happens when capital and land cannot move between sectors? Who wins and who loses from trade?

Discussion and Further Reading

Further reading:

- Ricardo, D. (1817), *On the Principles of Political Economy and Taxation*, Chapter 7.
- MacDougall, G.D.A. (1951), "British and American Exports," *Economic Journal*.
- Golub, S. & Hsieh, C.T. (2000), "Classical Ricardian Theory of Comparative Advantage," *Review of International Economics*.

Discussion

1. Should Canada be concerned about trade with Vietnam in textiles, given that Canada produces textiles? (Hint: what is Canada's comparative advantage?)
2. If Bangladesh's productivity doubles in every sector, how does this affect comparative advantage? Wages? Trade patterns?
3. Why does the Ricardian model predict complete specialisation but we observe partial specialisation in reality?